

Aircraft MRO: Chennault International Airport, Louisiana, USA — New Megadoor hangar door installation

# Megadoor vertical lifting fabric doors bring Chennault money-saving hurricane protection

Situated in Lake Charles, Louisiana, Chennault International Airport fills an important niche on the US Gulf Coast. The airport provides Maintenance, Repair and Overhaul (MRO) services for the aircraft of international companies, which has made it a valuable gateway for business in the local area.

Chennault's most prominent building, Hangar H, was opened in 2014. Covering an area of 112,000 ft$^2$ (10, 405 m$^2$) ,this high-capacity facility is occupied by AAR Aircraft Services, the largest MRO provider in the country. At both ends are 260' x 65' (80m x 20m) Megadoor vertical lifting fabric doors from ASSA ABLOY Entrance Systems, which were central to the hangar's construction.

## Customer challenges

Randy Robb, Executive Director of the Chennault International Airport Authority, first encountered Megadoor solutions at the ASSA ABLOY Entrance Systems booth at an annual MRO show. The idea for Hangar H was then on his mind, along with serious concerns about wind. In 2004, Hurricane Rita had caused $43 million dollars in damage to Chennault, including damage to five of six hangars that included their bottom rolling doors.

Robb was impressed by the Megadoor reference installations presented by ASSA ABLOY Entrance Systems, especially when it came to wind performance. The references were comprised of hurricane-proven hangars, including the US Coast Guard Hangar in Opa Locka, Florida, which has withstood multiple hurricanes without damage since its installation in 1998.

## What we provided

When Robb and his team decided to proceed with Hangar H, they took full advantage of the Megadoor properties. Wind speeds of 110 mph could easily be accommodated, because a Megadoor system can be designed for virtually any wind pressure – simply by varying the quantity and size of the horizontal beams. Whenever the door is fully closed, an automatic wind lock secures the building envelope, preventing building uplift of the sort that ravaged Chennault in 2004.

Hangar H was built in a drive-through configuration and sized to accommodate two massive C-5 cargo aircraft with T-tails. Thanks to the unique Megadoor construction, the door opening could be given a "stepped" design, instead of the rectangular design that would have been required for bottom rolling doors. Since no tracks are involved in a Megadoor system, it was also possible to utilize an existing concrete ramp from Chennault's days as a post-WWII US Air Force Base, despite a significant slope across the ramp opening.

## Benefits

Thanks to the Megadoor solution, Chennault International Airport achieved a hurricane-resistant hangar while saving a considerable amount of money in its construction. Because the slope of the existing ramp could be accommodated in the door design, there was no need to demolish and repour a significant amount of ramp – as would have been the case with bottom rolling doors.

Moreover, the stepped door design allowed for a dramatic reduction of the building's eave heights, which translated into substantial building cost savings.

Nonetheless, the new hangar offers full operational flexibility. With design support from ASSA ABLOY Entrance Systems, the stepped door was laid out such that aircraft of all types can be moved through the hangar and placed without constraint, which is a vital factor for productive MRO business.

## Testimonial

***"We needed to achieve the kind of flexibility that could be provided by the Megadoor concept in Hangar H to support the tenant's wide variety of aircraft – from single aisle to wide body. We worked directly with the ASSA ABLOY Entrance Systems team from the design phase through completion of the hangar. The results exceeded every metric we could envision. The entire evolution was smooth and we completed the project on time and on budget." – Randy Robb, Executive Director, Chennault International Airport Authority***

**Owner:** Chennault International Airport Authority
**Tennant:** AAR
**Engineer:** Meyer and Associates
**Contractor:** Alfred Palma
**Pre-engineered metal building supplier:** Nucor
**Hangar doors:** 260' x 65'5" (80 m x 20m) stepped Megadoor systems (2) in five parts

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